

Power BI Reporting & Business Analysis Project

SAKILA MOVIE RENTAL ANALYSIS

Excel | SQL | Power BI | DAX



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PROJECT OVERVIEW

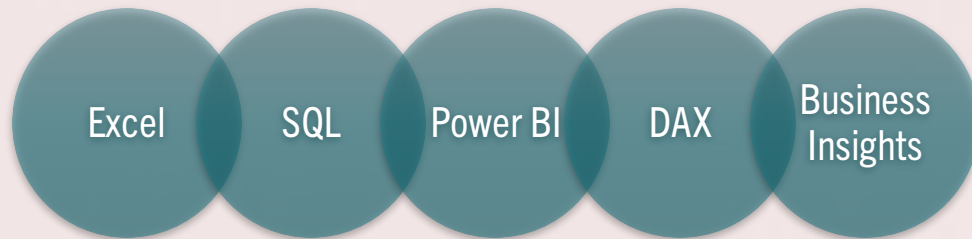
Objective

The objective of this project is to analyze the Sakila Movie Rental database and build an interactive Business Intelligence dashboard to uncover insights related to revenue, customer behavior, film performance, inventory trends, store operations, and geographic revenue distribution.

Business Goals

- Analyze rental revenue trends across different countries
- Understand customer purchasing behavior and payment preferences
- Identify top-performing film categories and inventory patterns
- Evaluate staff and store performance
- Generate actionable business insights using Power BI dashboards

Project Workflow

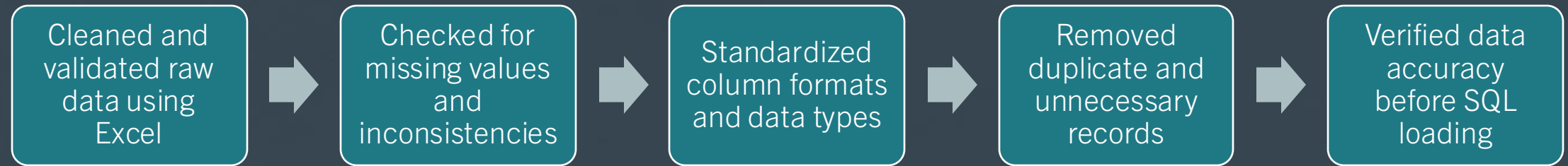


DATA CLEANING & PREPARATION

Data Source

The project uses the Sakila Movie Rental database containing customer, rental, payment, film, inventory, staff, and store-related data.

Data Cleaning Process



SQL Data Preparation

- Imported cleaned datasets into SQL Workbench
- Verified table relationships and data integrity
- Performed data validation checks using SQL queries

Raw Data → Excel Cleaning → SQL Loading → Validation → Power BI

DATA MODELING & DAX

Data Modeling

- Built table relationships in Power BI
- Connected customer, rental, payment, film, and inventory tables
- Created a structured relational data model

DAX Measures

- Total Revenue
- Total Rentals
- Total Customers
- Average Customer Rating
- Top Revenue Country
- Top Film Category

Outcome

Created dynamic KPI measures and an optimized data model for interactive dashboard reporting.

EXECUTIVE DASHBOARD

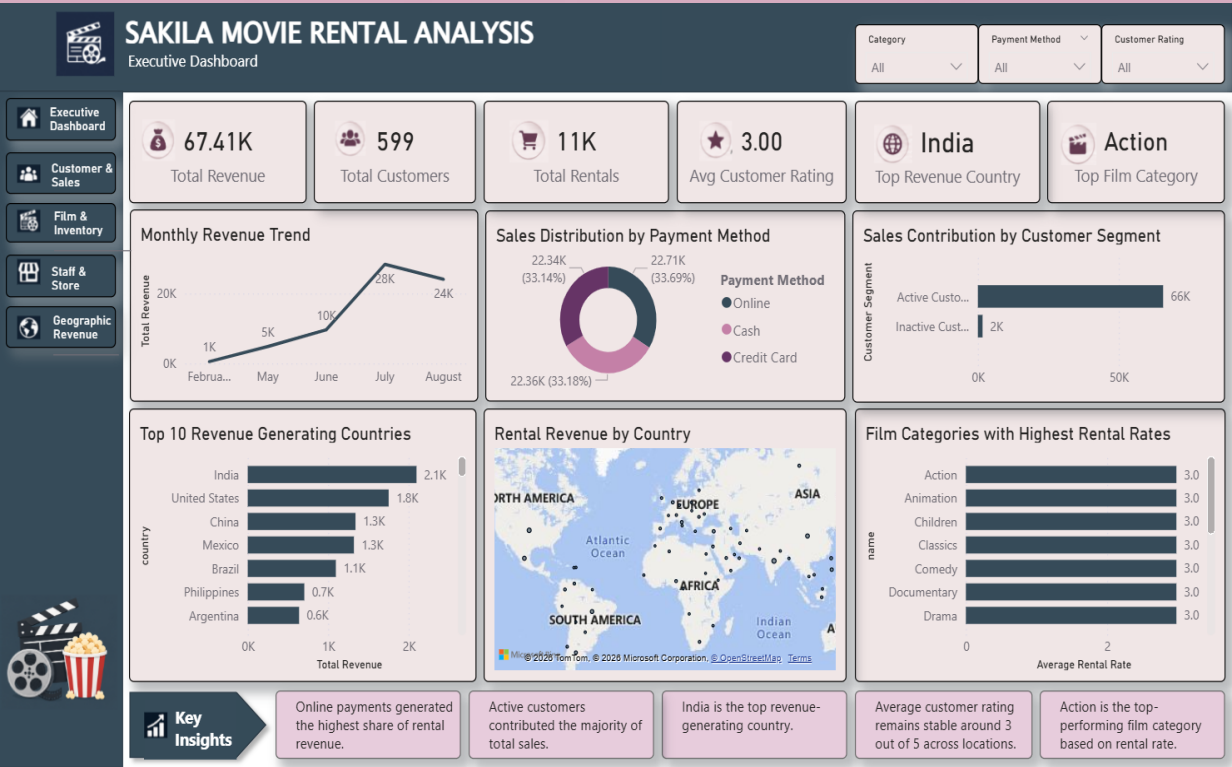
The Executive Dashboard provides a consolidated view of rental business performance through interactive KPIs, revenue trends, customer insights, payment analysis, and geographic revenue distribution.

Key Features

- Interactive slicers and page navigation
- Dynamic KPI tracking using DAX measures
- Revenue and customer trend analysis
- Geographic and category-based insights
- User-friendly executive reporting interface

Business Value

The dashboard enables quick decision-making by transforming raw rental data into meaningful business insights through interactive visual reporting.



CUSTOMER & SALES ANALYSIS

This section analyzes customer behavior, sales contribution, payment preferences, and customer rating trends to understand overall rental business performance.

Key Areas Analyzed

- Sales distribution by payment method
- Monthly revenue trends
- Customer segmentation analysis
- Customer ratings by location
- Customer distribution across cities

Key Insights

- Online payments generated the highest share of revenue
- Active customers contributed the majority of total sales
- Revenue showed strong growth during peak months
- Customer ratings remained stable across most locations

Business Value

Customer and sales analysis helps identify purchasing behavior, revenue-driving customer segments, and payment trends for better business decision-making.



This section analyzes film categories, rental duration, inventory distribution, and film ratings to evaluate inventory performance and rental patterns.

- Distribution of films by rental duration
- Inventory distribution by film rating
- Film categories in inventory
- Average rental duration by category
- Film categories with highest rental rates

- Action category showed strong rental performance
- Film inventory was distributed across multiple rating groups
- Rental duration remained consistent across categories
- Inventory distribution supported category diversity



STAFF & STORE PERFORMANCE

This section analyzes staff performance, store revenue contribution, employment distribution, and customer rating trends to evaluate operational efficiency.

Key Areas Analyzed

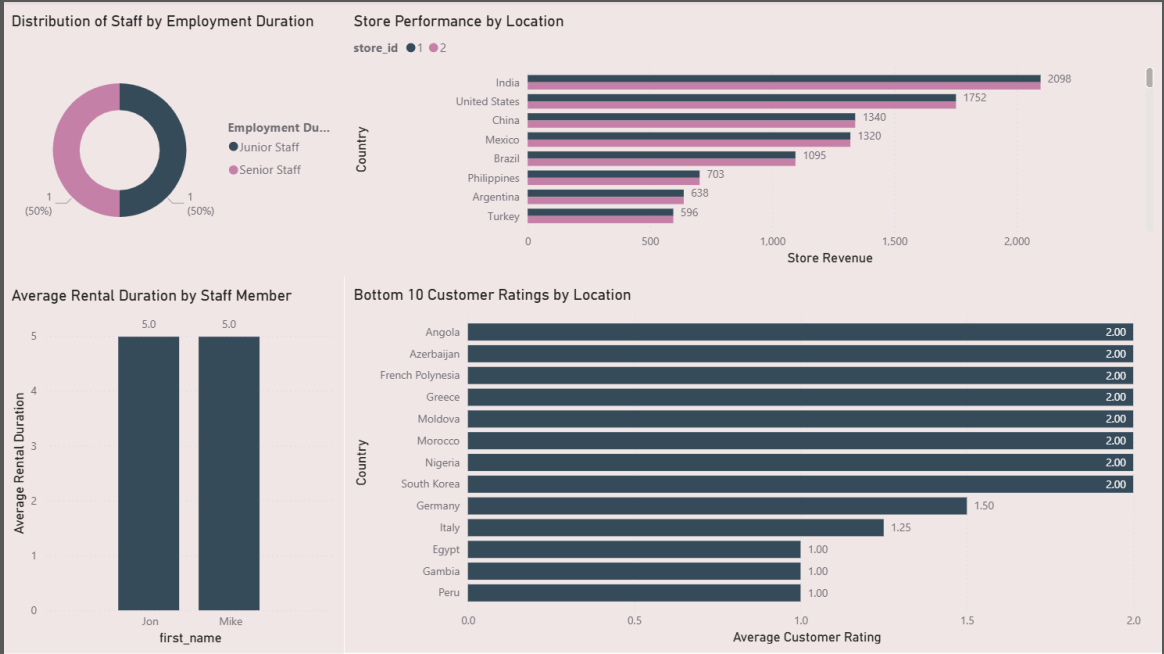
- Distribution of staff by employment duration
- Store performance by location
- Average rental duration by staff member
- Customer ratings across locations

Key Insights

- Store revenue varied across different countries
- Staff rental duration remained consistent
- Revenue contribution was higher in top-performing locations
- Customer ratings showed moderate variation across regions

Business Value

Staff and store performance analysis helps evaluate operational effectiveness, monitor store contribution, and support better business management decisions.



GEOGRAPHIC REVENUE ANALYSIS

This section analyzes country-wise revenue distribution to identify high-performing and low-performing geographic regions.

Key Areas Analyzed

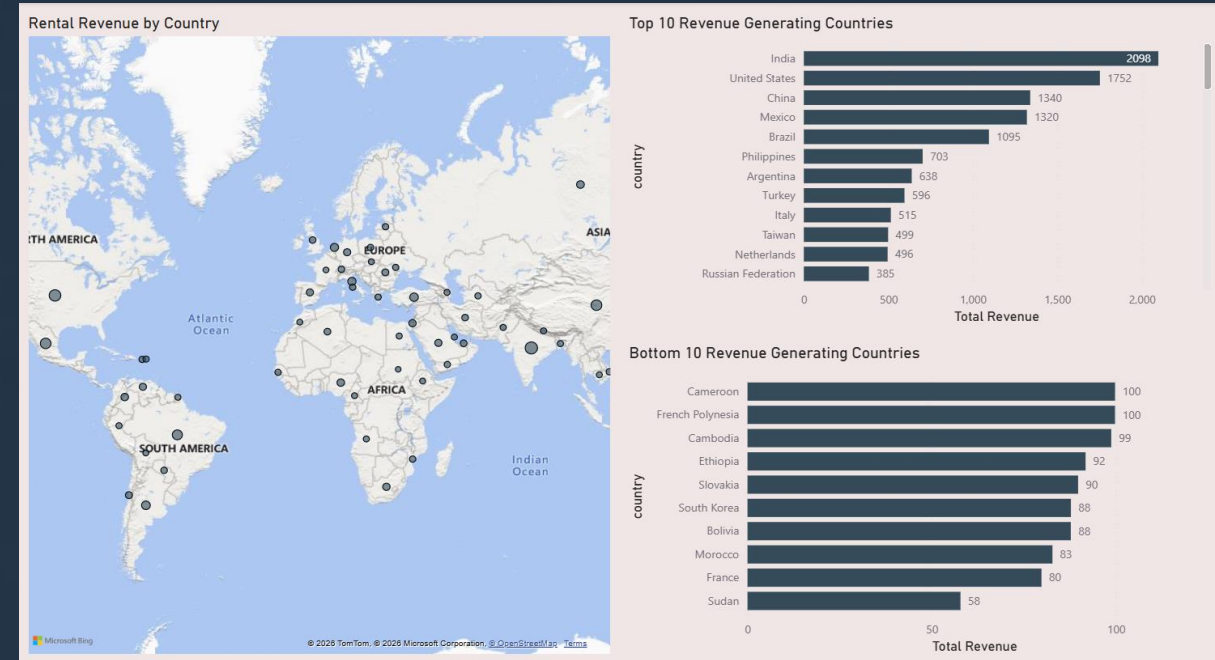
- Rental revenue by country
- Top revenue-generating countries
- Bottom revenue-generating countries
- Geographic revenue distribution trends

Key Insights

- India generated the highest rental revenue
- Revenue contribution varied significantly across countries
- Several regions showed low revenue performance
- Geographic analysis highlighted major revenue-driving markets

Business Value

Geographic revenue analysis helps identify profitable regions, monitor market performance, and support strategic business expansion decisions.



KEY INSIGHTS

India generated the highest rental revenue

Online payments contributed the highest sales share

Active customers drove most of the revenue

Action was the top-performing film category

Revenue increased significantly during peak months

BUSINESS FINDINGS

Geographic regions showed uneven revenue distribution

Customer behavior strongly impacted revenue trends

Payment preferences influenced sales performance

Inventory diversity supported rental demand

Interactive dashboards improved business analysis

CONCLUSION

The Sakila Movie Rental Analysis project successfully transformed raw rental data into meaningful business insights through interactive Power BI dashboards, KPI analysis, and data-driven reporting.

RECOMMENDATIONS

- Focus on high-performing geographic regions to maximize revenue
- Encourage digital payment methods to improve sales efficiency
- Expand inventory for top-performing film categories
- Improve engagement strategies for inactive customers
- Use dashboard insights for continuous business monitoring and decision-making

THANK YOU

Presenter

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